Predictive Model Plan – Geldium Delinquency Dataset

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1. Model Logic

Proposed model: Logistic Regression (simple, interpretable) and Random Forest (complex, higher accuracy).
Chosen model: Logistic Regression for transparency and regulatory compliance.
Key input features: Credit Utilization, Credit Score, Missed Payments, Debt-to-Income Ratio, Employment Status.
Workflow: Data ingestion → Feature selection → Model training → Prediction output (delinquency risk score).

2. Justification for Model Choice

Logistic Regression is interpretable, easy to deploy, and widely used in financial risk modeling. It provides clear coefficients showing how each variable impacts delinquency risk. This transparency ensures compliance with regulatory requirements for financial decision-making. While Random Forest may offer higher accuracy, its complexity and lack of interpretability make Logistic Regression a better fit for Geldium's operational and compliance needs.

3. Evaluation Strategy

Metrics: Accuracy, Precision, Recall, F1 Score, AUC.
Fairness checks: Compare performance across demographic groups (age, employment status, location).
Bias mitigation: Re-sampling, fairness-aware algorithms, and monitoring feature importance.
Ethical considerations: Avoid discriminatory predictions, ensure transparency in customer communication.